

Conducting Spreadsheet Audits as a (re)Design Method: A Bird Biodiversity Decision Support Tool Case Study

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Abstract

Farms are agro-ecological systems, where farmers work to balance agronomic, economic, and ecological outcomes. Digital Decision Support Tools (DST) assist farmers and their collaborators in navigating complex choices. DSTs are commonly developed by agricultural domain experts, and implemented as customized spreadsheet tools that contain a blend of curated data and expert decision logic. With increasing availability of high-quality data, improved scientific understanding of the relationships among agricultural decisions, and demand for more user-friendly tools, there is interest in transitioning existing spreadsheet tools to more dynamic software. In this paper, we describe the first phase of a collaborative effort to redesign a Microsoft Excel spreadsheet tool developed by the Wild Farm Alliance, to support bird habitat and improve on-farm biodiversity. We present this case study to the HCI community to provide actionable guidance and recommendations on how to conduct an Excel spreadsheet audit as part of a software (re)design process.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**.

Keywords

case study, user-centered design, sustainability, decision support tools, spreadsheets, design methods

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1 Introduction

Almost 50% of the United States' 2.26 billion acres of land are dedicated to agriculture, through both range lands or crops land, while a mere 14% are designated as parks and wildlife areas [28]. The conversion of natural ecosystems to agriculture lands continues to negatively impact biodiversity and habitat availability, air, land, and water quality and quantity, and to contribute to climate change

through greenhouse gas emissions [12]. Recognizing conservation opportunities on land outside of protected parks, such as farmland, is critical for preserving wildlife and supporting biodiversity. This is especially important for birds who have lost 30% of their North American population since 1970 [24]. Agricultural sustainability poses an increasingly complex decision landscape as farmers balance short-term goals, (e.g., yielding a marketable crop), intermediate experimentation with new practices (e.g., reducing tillage to improve soil health outcomes), and implementation of longer term strategies with broader social, ecological, and agricultural outcomes (e.g., redesigning landscapes to create pollinator habitats).

Farmers and their advisors collaborate to determine effective strategies to improve farm sustainability outcomes, reflecting and analyzing the current state of a farm - site, structure, context, and practices, to identify areas for sustainable improvement [25]. Farm advisors can range from university researchers, industry representatives, government agency representatives, and members of non-profit advocacy organizations. Such advisors are typically subject matter experts and have historically created and curated information products and decision making aids to identify and implement new farm management strategies, including conservation practices that can improve biodiversity outcomes.

Decision Support Tools (DSTs) have emerged as a popular genre of agricultural technologies, designed to assist farmers and their advisors in semi-structured decision making via codified expert guidance, practice recommendations, and/or analytics [29]. Adoption of DSTs remains low with barriers to adoption ranging from: low software quality (e.g., usability, ease of use, reliability) [8], lack of suitability for whole-farm management (versus supporting a single decision points like irrigation timing) [29], overly generic recommendations [23], and a lack of training for farmers and their advisors alike [20]. While the goal of improving access to actionable conservation knowledge, including via decision support, is colloquially and commonly promoted in the agriculture community, a relationship between software design improvements and impacts on biodiversity adoption long term has yet to be empirically characterized or demonstrated. Similarly, while significant attention has been paid to characterizing barriers to technology adoption, there remains a lack of design guidance on how to remedy the software quality challenges as diverse agricultural experts pursue the design of DSTs.

Agricultural DSTs are often initially prototyped or implemented in spreadsheet software such as Microsoft Excel, as it is familiar to users, easy to share, and has reduced development needs [6] [16] [19] [22]. However, as projects grow in complexity Excel can impose significant limitations on a tool's continued development [6] [16]

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[17]. While Excel's base functionality can be expanded through Visual Basic for Applications (VBA) macros and third-party packages, once engineering reaches this level, it is often more efficient and interpretable to transition to advanced programming environments [16] [17]. These environments offer superior statistical capabilities, faster processing speeds, version control, debugging, modifiability, improved transparency, and better collaboration options. Thus, converting spreadsheet DSTs to interactive software applications is a common project in agricultural technology research and development.

Despite continued interest in transitioning DSTs from simple, static, spreadsheet tools to complex, interactive, web applications that integrate data across geographies, ecologies, and agronomic systems, there is limited research documenting the evolutionary design and redesign process. This paper describes some of the emergent processes underlying our human-centered approach to (re)designing DSTs, where we collaborate with agriculture, conservation, and informatics experts. Specifically, we will share a case study, where we use human-computer interaction (HCI) methods to "audit" an existing conservation planning spreadsheet DST designed to promote adoption of conservation practices that support bird biodiversity in Californian agriculture.

2 The Case Study: Wild Farm Alliance & the Beneficial Bird Habitat Assessment & Native Plant Tool

The Wild Farm Alliance (WFA) is a national agricultural and conservation non-profit with a two-fold vision to support "the conservation of wild nature and the sustainability of farming systems" [5]. Since 2000, they have worked to improve biodiversity outcomes and ecosystem health by re-integrating 'wild' habitats in rural and agricultural landscapes [5] by providing specific guidance, educational programs, information resources, as well as decision support on how to identify and implement appropriate conservation practices that promote biodiversity [2]. These practices can range from providing simple nesting boxes and/or perches for birds, creating hedgerows or windbreaks on-farm borderlands, to creating more complex on-farm habitats.

WFA developed a spreadsheet DST, the "Beneficial Bird Habitat Assessment & Native Plant Tool" (the 'WFA Bird Tool', in short) [2], to help Californian farmers engage in bird friendly habitat and management practices. This work was funded by the U.S. Department of Agriculture's Natural Resources Conservation Service (NRCS) and a range of conservation-oriented organizations. These practices, such as creating riparian zones, support bird populations, who in turn provide pest management and other services to farmers. This spreadsheet DST consists of two primary components. First, a **habitat assessment module** that walks the user through a farm and land management questionnaire to create a 'bird score' reflecting the impacts of current and proposed practices on biodiversity outcomes. Second, a **plant selection module** that evaluates and recommends suitable native plants to create bird habitats specific to a user's context.

The WFA Bird Tool was initially developed in-house as a Excel spreadsheet, inspired by the Xerces Society's habitat assessment tool—a PDF-based spreadsheet designed to support invertebrate

biodiversity. After receiving positive feedback on the original tool the WFA decided to transition to a web-based platform in order to advance the tool's functionality, increase access, and expand its applicable range beyond California. The WFA engaged Purdue's Agricultural Informatics lab (Axilab) [1] to conduct design research, create a design specification, and support subsequent implementation of DST. Our research lab has converted and expanded other agricultural DST spreadsheet tools in the past, notably a cover crop decision tool [4]. Through this work, we have used HCI design methods to help assess these tools and their information infrastructure and create development plans. In this case study we will present our strategy and key takeaways in hopes they may be used as a road map for redesigning other agricultural tools.

3 Strategy: Spreadsheet Audits as a Design Method

Before work on the new tool could begin, we needed to conduct a comprehensive audit of the original spreadsheet-based WFA Bird Tool. Spreadsheet audits are commonly performed during quality assurance and control processes, as well as for debugging and implementing minor updates in order to discover the source of errors, improve understanding of data dependencies, data flows, and to align tools with user needs [9, 10, 13]. In this case study we use an analogous audit process as the first step in our redesign effort. That is, our audit was used to establish a baseline of the structure and behavior of the current tool, in order to prevent unnecessary overhauls, direct new pathways and extract domain knowledge [15]. No holistic and clearly applicable audit methodology was available, thus we describe our emergent method in the sections that follow.

During the audit, our team organized and documented important information, created materials for the future development team, and identified opportunities for improvement. Gathering, documenting, and synthesizing this information has proven to be key to facilitating clear communication and collaborative road mapping between the "design team", i.e. the researchers at Axilab, and the "clients", i.e. the WFA team that provided funding, leadership, and subject matter expertise.

Two key questions guide our audit. Firstly, what is the tool's intended function i.e. what are the requirements? The tool's requirements are a fundamental consideration that should be identified and understood at the outset during initial client meetings. Secondly, how is the tool fulfilling (or not fulfilling) its requirements i.e. what is the system architecture? In answering this question Excel's constraints often become apparent as evaluations of the tools performance differ from the clients' original description of its requirements. These discrepancies highlight the compromises and workarounds Excel can implicitly impose on a tool, and can be used to target improvements [17].

User satisfaction with a DST is primarily determined by information quality (e.g., accuracy, completeness, etc.) and system quality (e.g., ease of use, reliability, etc.) [7]. Therefore, we focus on two core areas of system architecture during the audit: data and functionality. Ensuring robust data sources and practices are central to any DST, as the data creates and justifies the support given to users. Depending on the scope of the project, the redesign team must understand issues such as data provenance, data acquisition,

data transformation, and data models. Functionality encompasses the way the data is used in the tool, such as architecture, logic, performance, and user interaction. The functionality encapsulates the expertise a DST offers its users. It should accurately synthesize and reflect domain knowledge and provide users actionable and constructive support.

4 Emergent Design Methods

In this section, we describe a suite of design methods that were created and/or adapted to conduct the spreadsheet audit of the WFA Bird Tool: document gathering, data exploration, stakeholder interviews, user journey mapping, and process logic mapping (Figure 1). Throughout the audit the team iteratively and collaboratively produced data dictionaries, schematics, and other information artifacts to inform meetings with stakeholders and help guide future development plans. The iterative process of exploration, mapping, and discussion resulted in a nuanced understanding of the tool's requirements, system architecture, data and domain application, and uncovered both design challenges and opportunities.

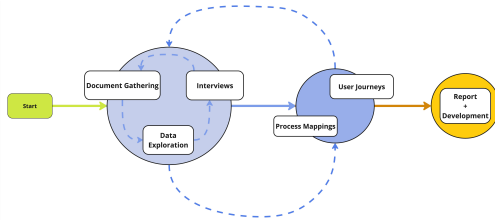


Figure 1: Map of the iterative process through five design methods: document gathering, data exploration, stakeholder interviews, user journey mapping, and process logic mapping. User journeys and process mappings, collaboratively derived from data exploration and interviews, represent key intermediate steps toward the final report and development plan. This cyclical approach emphasizes the integration of insights at each stage to inform the evolving process.

4.1 Document Gathering

Gathering documentation for review is the first step in our audit process. This includes requesting internal and public materials, unlocked versions of the tools, and a list of nearest-neighbor tools. These documents facilitate the next steps of the audit and, in turn, guide stakeholder interviews. Ensuring that the design team has gathered a full set of documentation from the beginning streamlines the audit and minimizes requests for information. Table 1 details the document types requested and the key insights gleaned from each.

4.2 Data Exploration

Data exploration encompasses a variety of steps, such as document reviews, tool testing, and validation, that are used to create a data inventory, synthesize metadata and/or relevant contextual information, and create a data model. These artifacts allow for analysis of the information quality and the data pipeline and uncover how the

data are used and manipulated through the tool to create recommendations and other outputs. Beginning data exploration before interviews allowed the design team to identify lines of inquiry, areas in need of clarification, and ensured that we entered interviews with a foundational understanding of how the data influence tool scope and functionality, as well as the user experience.

Hendry et al. summarizes "the strengths [of spreadsheet tools] are such as allow quick gratification of immediate needs, while the weaknesses are such as make subsequent debugging and interpretation difficult" [14]. These debugging and interpretation issues make decomposing the 'backend' of excel tools difficult and time consuming, especially for those unfamiliar with them. Many important statistical methods are not obvious on first inspection. Cells may be hidden, 'very hidden', collapsed, or intentionally placed out of view. VBAs/macros must be explicitly searched for and can be idiosyncratic. Functions can be difficult to decipher when values are nested and spread across multiple sheets, which is necessary for many complex tools; "cell chasing", or the process of tracking these values down, is slow and prone to error [26]. Additionally, some features are not available on Unix operating systems and will be hidden to users on these systems. As a result, understanding and validating Excel tools can become extremely complex and time-consuming, even when the tool performs a relatively easy task [14] [16] [26]. For teams prototyping their tool in Excel, having strong documentation helps facilitate these processes. Table 2 describes the spreadsheet elements to identify and document.

WFA data exploration resulted in the following key outputs: data dictionaries, a set of input variables, a demo relational database, and a data model. The readability and transparency issues, as outlined above, were one of the main roadblocks in our Excel audit. By condensing elements of the tool into descriptive artifacts, such as those listed above, we created clarity around otherwise opaque processes. Through these activities, the redesign team developed a strong understanding of the tool while also producing artifacts to help direct the future development team in transitioning and preserving information between tools. While the bulk of work on these outputs was done during data exploration, they were not completed until interviews and cognitive walkthroughs (CW) were conducted.

These outputs are now guiding continued development of the tool. Data dictionaries, "a centralized repository of information about data descriptions such as meaning, relationships to other data, responsibility, origin, usage, and format. [Are] a basic tool within the database environment that assists company management, database administrators, systems analysts, and application programmers in effectively planning, controlling, and evaluating the collection, storage, and use of the data resource" [27]. The data dictionary created during the audit is now being used to manage our data expansion efforts in the Midwest and Eastern states.

Additionally, we are conducting qualitative research with domain experts, such as ornithologists and conservation workers, to ensure that the scores assigned to conservation practices, in the habitat assessment module questionnaire, are regionally appropriate as we expand. The original rationale was based in California, reflecting local values that may vary in new Midwest and Eastern states. For instance, the valuation of water related conservation practices may differ in drought prone areas or places critical to waterfowl.

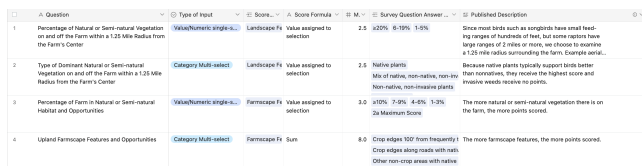
Table 1: WFA Bird Tool: Documents Gathered

Category	Documents	Insights
The WFA Bird Tool	Unlocked version of the Excel tool	Allowed the team to decompose and play with the tool.
Internal Documentation	Data Sources and Contact Sheets, Raw Data Files, Intermediate Data Processing Files, Annotated ‘under the hood’ versions of the tool	Outlined the data pipeline from source to final implementation. The annotated version of the tool stated the goal of various components, helping explain functions later on when mapping the process logic.
Public Documentation	WFA Habitat Assessment Guide with Appendices, Webinars	These stated the clients objectives, provided domain background, and helped clarify the user base. The appendices were especially useful in tracing data through the pipeline.
Nearest Neighbor Tools	A list of tools the WFA was inspired by.	Helped identify WFA goals.

Table 2: Key Elements to Document

Element	Rationale
Formulas and values	Reduces cell chasing.
User-defined functions	Improves validation and accelerates exploration.
Hidden and collapsed columns	Prevents oversight.
External data connections	Improves validation and accelerates exploration.
Macros	Improves validation and accelerates exploration.
Pivot tables	Improves validation and accelerates exploration.

By decomposing the questionnaire into an assessment matrix we improved readability and condensed important information across various documents such as question and answer options, score formulas and weights, domain rationale, and metadata (Figure 2). This product was then used to design our future qualitative study and the survey we are providing to domain experts.



Question	Type of Input	Score Formula	Score	Published Description
1. Percentage of Natural or Semi-natural vegetation on and off the Farm within a 125 mile Radius from the Farm's Center	Value/Numeric/Single-Select	Landscapes FI Value assigned to selection	0-5 (0-25%) 6-10 (5-25%)	Since most birds such as songbirds have small feeding ranges of hundreds of feet, but some species have larger ranges of 1 mile or more, we chose to measure a 125 mile radius surrounding the farm. Example aerial...
2. Type of Dominant Natural or Semi-natural Vegetation on and off the Farm within a 125 mile Radius from the Farm's Center	Category/Multi-select	Landscapes FI Value assigned to selection	1-5 (Native plants: 1-3, Non-native, non-invasive plants: 4-5)	Because native plants typically support birds better than non-natives, they receive the highest score and invasive weeds receive no points.
3. Percentage of Farm in Natural or Semi-natural Habitat and Opportunities	Value/Numeric/Single-Select	Farmhouse FI Value assigned to selection	0-10 (0-25%, 4-10%, 5-25%)	The more natural or semi-natural vegetation there is on the farm, the more points scored.
4. Upland Farmhouse Features and Opportunities	Category/Multi-select	Farmhouse FI - Sum	0-10 (Crop edges 100% from frequency) 1-5 (Crop edges along roads with trees) 6-10 (Other non-crop areas with native)	The more farmhouse features, the more points scored.

Figure 2: Snapshot of the assessment matrix of the Bird Tool questionnaire. This document captured data from the front-end of the Excel tool, hidden sheets and columns, internal documentation, and the published tool appendix.

A benefit of working from an Excel tool over starting from scratch is that the data has already been functionally organized - "in the very process of laying out data in a spreadsheet, the user is viewing and studying the data" [19]. The existing Excel structure of sheets, columns, rows, and cells contains important information on data relationships, domain knowledge, and tool requirements [15]. This tacit knowledge facilitates the data modeling and management work as there is a baseline system to build from.

While the existing tool may have implicit data relationships, advanced programming techniques that make these connections and structures explicit are uncommon. Excel provides some capabilities for linking data, but these methods are restricted and rarely used in these types of DST as Excel is often chosen explicitly for quick prototyping of task-specific tools and lack of programming skills [19]. In our case the tool had no inherent data structure linking information across the tool. Instead, ‘look up’ functions were used in VBAs. These ‘look up’ functions and other data linking techniques comprised the bulk of our tool’s programming, demonstrating Excel’s tendency to over complicate tasks. Using our understanding of the WFA’s data, developed through data exploration, process mapping, and subsequent interviews, we moved the data into Airtable. We structured this data via entity-relationship diagrams that depict the underlying data schema (Figure 3). This work will serve as a blueprint as we move forward with tool development.

4.3 User Journey Mapping

User journey maps are used to illustrate screen flow, activities and subroutines, personas/profiles, and outline system requirements [11]. In Excel tools, activities are often grouped into tables or sheets, buttons may be used to trigger scripts and functions, and navigation is directed through text boxes or triggered by buttons. Transitioning from Excel to a new interface inherently changes a tool’s UI/UX. Therefore, design details, such as where a button is or what it looks like, are not a focal point. Rather, user journeys explored user-facing logic and processes, and contextualized back-end processes.

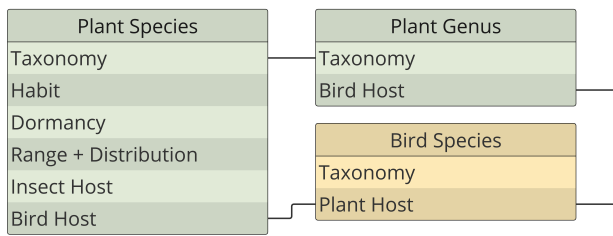


Figure 3: Simplified entity-relationship diagram based on WFA data, illustrating the connections between bird and plant classes in the new database. The diagram highlights clusters representing 44 plant species, 11 plant genera, and 17 bird species variables.

By testing the tool using scenario-based evaluations or personas, the redesign team may explore these elements and map the user experience with the tool.

We created drafts of our user journeys in Miro, utilizing screenshots of the WFA tool and annotations (e.g., arrows, sticky notes) to document user progress across sheets (Figure 4). Subsequently, these drafts were streamlined and finalized for presentation. The user journey process helped delineate the tool's distinct modules, their components, and respective requirements. This clarity facilitated future process mapping by categorizing data and processes into their corresponding modules.

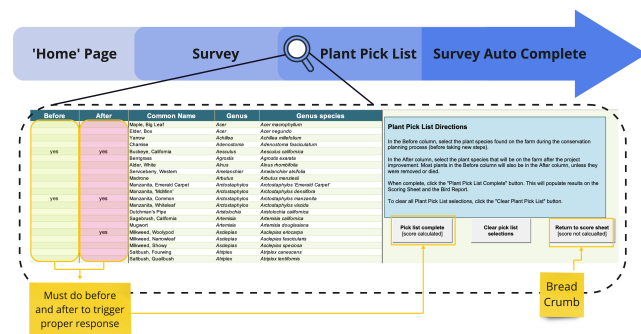


Figure 4: A simplified diagram of the user's journey through the Excel tool. The 'magnified' section, outlined by the dotted line, captures a small piece of the actual user journey in Miro. This expert depicts a section of the plant pick list module.

In addition, the user journey illuminated ambiguous or difficult interactions. For example, one question asked users to indicate the "Percentage of Natural or Semi-natural Vegetation on and off the Farm within a 1.25 Mile Radius from the Farm's Center". This question required users to undergo a clunky process: leaving Excel, accessing a mapping tool, outlining a 1.25-mile radius, and estimating land cover. Finally, journey mapping also helped identify the implicit applications of domain knowledge. For example, the decision to group questionnaire sections into categories - landscape, farmscape, and management - and the weights assigned to each

category reflected the WFA's perspective on the value of different conservation and agricultural principles.

4.4 Process Logic Mapping

Process logic mappings are typically used to outline complex procedures and sequences [21]. In the audit, we applied them to data pipelines, backend functionality, and client systems. In Excel audits these mappings help untangle and document 'cell chasing' and other backend engineering methods, improving transparency. Additionally, missing technical documentation, see Table 2, is also collated and finalized during this method.

In our case study, we were tasked with expanding the WFA database beyond California. Accordingly, we also created process logic maps for the entire WFA data pipeline, from source to implementation. While the internal documents and the published guide described most of their data collection and cleaning process, some gaps remained. By creating process mappings during our interviews, we were able to better understand and document exactly how they acquired and prepared their data and what data items would need to be replicated going forward (Figure 5). We created these maps in Miro using screenshots, embedded recordings, and annotations.

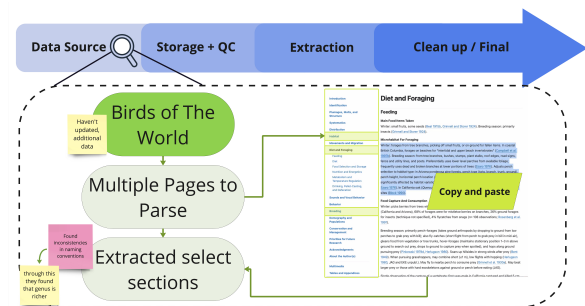


Figure 5: A simplified process map created during a stakeholder interview, illustrating the WFA data acquisition workflow. The zoomed-in section highlights the initial steps in sourcing bird-plant relationship data.

4.5 Stakeholder Interviews

Interviews and CW serve to gather detailed insights on objectives and functionality of the tool, as well as to address gaps identified during the data exploration process. Combining both methods ensures comprehensive and in-depth discussions. According to Mahatody et al. (2010), CW should be directed by "forms with specific questions that must be answered [to] guide the user through the walkthrough. CW can be applied in the system design and development phase to identify system usability problems as soon as the system is modeled and its features specified. It can also be used retroactively to determine the difficulties in using a system through executing specific scenarios and special testing" [18].

Following the data exploration stage of our audit, many details remained unclear, such as how data was being linked, technical details of the data pipeline, and domain knowledge behind certain questions and processes. These questions were condensed into

Table 3: Interview and Cognitive Walkthrough Insights

Category	Description	Insights
Domain Walk-through	Walked through the tool, focusing on the rationale behind questions, responses, and scoring systems.	Defined the user base as two cohorts: technical (farm advisors) and non-technical (farmers, and landowners). Documented domain knowledge influencing category/question inclusion and scoring weights, outlining the underlying rationale in the assessment.
Technical Walk-through	Discussed processing logic, including tool functions and VBA code.	Identified hidden cells containing critical formulae. Deconstructed and analyzed VBA subroutines. Formalized data schema and representation structures. Validated data exploration insights regarding sheet dependencies and function mappings.
Data Pipeline	Reviewed the guide, appendices, and internal files used for data storage and processing before tool usage.	Outlined each procedural step, including keyword-based data point generation and the exception handling logic (e.g., plurals, aliases, etc.). Identified bottlenecks in the original methodology, such as cell limits and reliance on manual methods.
Final Interview	Presented finalized mappings and findings	Confirmed findings and provided a last opportunity edits.

interview scripts, following Mahatody’s method, and supplemented with user journeys and process mappings. These scripts were then used to conduct semi-structured interviews with our participants. The interviews were reviewed to increase the fidelity of the process maps and other design artifacts presented in this document.

We conducted separate CW for the front and back end to allow for in-depth exploration of each. This approach ensured participants could focus on the areas most relevant to their expertise. Additionally, we created live process mappings during the CW to capture complex processes. We used Miro, a collaborative design platform, to make our mappings and to allow stakeholders to make direct annotations.

Interviews and walkthroughs were held over Zoom with WFA’s executive director Jo Ann Baumgartner, deputy director Shelly Conner, and Excel developer David Ponder. Interviews were recorded and mappings and notes were taken for our records. Our technical questions were addressed by David Ponder while interviews with WFA directors provided a deeper understanding of the data pipeline (Figure 5) and how domain knowledge influenced design and logic. Table 3 outlines our stakeholder interview timeline and the corresponding insights.

5 Conclusions

WFA first partnered with Axilab in 2023, to conduct the Excel audit. Since completing this work the audit was used to help secure a million dollar Conservation Innovation Grant through the NRCS [3], in partnership with industry and non-profit partners, that will continue to fund direct future development. After executing each of the five design methods – document gathering, data exploration, stakeholder interviews, user journey mapping, and process logic – a finalized report was created to organize and synthesize our work and present recommendations for the next version of the Bird Tool. During this initial redesign stage the audit facilitated comprehensive, clear, and precise discussion with the WFA and the iterative co-creation of a development plan. In our ongoing and future work,

we seek to demonstrate how software design improvements can better improve technology and conservation adoption, as well as produce improved DSTs for sustainable agriculture.

The goal of agricultural DSTs are to assist farmers in navigating decision making by analyzing data and sharing expert knowledge. To be effective, these tools must present this information in an accessible, user-friendly manner. In the case of the WFA’s Bird Tool, data and expert opinions are captured and shared in the two modules. First, the questionnaire incorporates a curated set of questions, along with associated weights and scores, reflecting expert opinions on the value of different conservation and agricultural practices. Second, the plant selection module compiles plant data and assigns cumulative points reflecting expert opinion on supportive bird habitats and plants.

For these modules to be successful, the underlying information quality must be robust, and the analysis and presentation of this information must be clear and understandable. By conducting an HCI driven audit, we prioritized a farmer-centered approach in every step. Iterative analyses and stakeholders meetings solidified the user, farmers and their advisors, as a guiding star. Centering their needs, we collaboratively identified future information requirements for improved user experience. These development efforts, such as regional expansion, are being directly supported by the audit artifacts: data dictionary, assessment matrix, etc. For example, the data dictionary is enabling us to seamlessly replicate and expand the plant module’s data, while the Airtable database is facilitating exploration of new analysis techniques and information presentation methods. Ultimately, these efforts will result in more actionable, accessible, and usable support for farmers engaging in bird conservation.

These development directions are being eased by our transition away from Excel to more flexible and robust environments. While Excel is a strong prototyping tool, the complexity of diverse regional ecological and agricultural data surpasses its capacity. The improved version of the Bird Tool, unshackled by the limitations of

a spreadsheet, will enhance usability and effectiveness, ultimately better supporting farmers and birds. The five methods used in this audit may be applied by other teams transitioning from Excel to more advanced systems by centering the user and the client, fostering communication and understanding, and directing the creation of critical documentation.

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